

Weekly Assignment: Division into Cases, Quotient-Remainder, & Floor/Ceiling (Epp 4.5 & 4.6)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 4.5 #17). Prove directly from the definitions that for every integer n , $n^2 - n + 3$ is odd. Use division into two cases: n is even and n is odd.

Problem 2 (Epp 4.5 #18 a). Prove that the product of any two consecutive integers is even.

Problem 3 (Epp 4.5 #40). Prove that if n is any odd integer, then $n^4 \bmod 16 = 1$.

Hint: Recall that an odd number can be written as $n = 2k + 1$ for some integer k . Expand $(2k + 1)^4$ and simplify modulo 16.

Problem 4 (Epp 4.6 #11). State a necessary and sufficient condition for the floor of a real number to equal that number.

Problem 5 (Proof: Divisibility by 3). Prove the following well-known divisibility rule from grade school:

If the sum of the digits of a (positive) integer is divisible by 3, then that integer is divisible by 3.

Hints:

1. A number written in decimal is a sum of its digits multiplied by powers of 10. That is, a number with digits $d_k d_{k-1} \cdots d_1 d_0$ equals

$$d_k \cdot 10^k + d_{k-1} \cdot 10^{k-1} + \cdots + d_1 \cdot 10 + d_0.$$

2. Recall that n is divisible by 3 when $n \bmod 3 = 0$.
3. The following properties of modular arithmetic may be useful:

$$\begin{aligned}(n + m) \bmod q &= ((n \bmod q) + (m \bmod q)) \bmod q, \\ (n \cdot m) \bmod q &= ((n \bmod q) \cdot (m \bmod q)) \bmod q.\end{aligned}$$

4. Observe that $10 \bmod 3 = 1$, so $10^j \bmod 3 = 1$ for every $j \geq 0$.